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Beyond species richness: Habitat fragmentation reduces community occupancy and functional richness of mountain forest mammals

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ABSTRACT

Forest fragmentation has critical and ongoing impacts on multiple facets of biodiversity. Yet, the specific impacts of fragmentation on animal communities, especially beyond species richness, remain less understood. Here, we addressed the question of how forest fragmentation affects both community occupancy (i.e., average occupancy across the species community) and functional richness of 39 medium- and large-sized mammals, accounting for imperfect detection and collinearity among landscape metrics. We conducted an intensive camera-trapping survey (731 camera traps, 79,108 trap nights) across a broad gradient of forest fragmentation in Southwest China. We investigated the relationships between fragmentation (quantified as patch density) and mammalian community occupancy, species richness, and functional richness for medium- to large-bodied mammals. We applied Bayesian hierarchical multispecies occupancy models to account for imperfect detection and assess both individual species and community-level responses. Additionally, we used piecewise structural equation modeling to disentangle the direct effects of fragmentation from those of forest cover, addressing collinearity among landscape metrics. Our results demonstrate that increased forest fragmentation significantly reduces both community occupancy and functional richness. Notably, functional richness continually decreased even when species richness leveled out along gradients of fragmentation, indicating that forest fragmentation filters for species that are more functionally similar, thus homogenizing functional diversity. Structural equation modeling consistently revealed forest fragmentation as a stronger determinant of both species and functional richness compared to forest cover, underscoring the importance of habitat configuration in determining species distributions and functional diversity of forest-dwelling mammals. At the species level, 85% of mammals exhibited a negative response to fragmentation, with nocturnal species and those with narrower habitat width being particularly sensitive. We conclude that forest fragmentation fundamentally impacts not just where species occur, but also the functional traits present within a community. Our findings highlight the importance of retaining large, contiguous forest areas in mountain landscapes for supporting diverse mammalian communities.

1. Introduction

Forest fragmentation is pervasive in both natural and human-modified landscapes, with 80% of the world's remaining forest patches smaller than 5 km^2 and 50% under 1 km^2 (Ibisch et al., 2016). It is recognized as a major driver of biodiversity loss (Ma et al., 2023; Tao et al., 2024), contributing to 20–75% reductions in species diversity across taxonomic groups, including plants, birds, and invertebrates (Haddad et al., 2015). By altering the size, shape, configuration, and connectivity of forest patches (Fahrig, 2003; Fletcher et al., 2018), fragmentation not only changes species diversity and community composition (Fahrig, 2003; Kuipers et al., 2021; Gonçalves-Souza et al., 2025) but also disrupts functional trait diversity crucial for maintaining ecosystem services (Rybicki et al., 2020; Yu et al., 2024). Moreover, fragmentation jeopardizes species interactions in natural habitats, affecting entire ecological networks (Hagen et al., 2012; Rybicki et al., 2020; Zhang et al., 2023a,b). Understanding and mitigating the impacts of forest fragmentation on biodiversity has become a central focus of

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ecology and conservation biology (Betts et al., 2019; Chase et al., 2020; Penjor et al., 2022).

While habitat fragmentation often arises from habitat loss in landscapes (Püttker et al., 2020; Valente et al., 2023a,b), its effects on biodiversity remain intensely debated (Miller-Rushing et al., 2019; Fletcher Jr. et al., 2023; Gonçalves-Souza et al., 2025). Driven by the goal of improving conservation management, the "Single Large or Several Small" (SLOSS) debate has centered on whether habitat loss or fragmentation poses the greater threat to biodiversity (Fahrig, 2017). For many years, the "habitat patch hypothesis" assumed that fragmentation negatively impacts biodiversity unidirectionally, largely by extending species–area relationship (SAR) patterns to fragmented landscapes (Ewers and Didham, 2006; Matthews et al., 2016). More recently, the "habitat amount hypothesis" (HAH) challenged this view, proposing that the total amount of habitat surrounding a site—rather than patch size or isolation—primarily determines local species richness (Fahrig, 2013). According to HAH, fragmentation per se contributes little to biodiversity change beyond its influence on overall habitat area (Fahrig, 2013). In a review of empirical studies, Fahrig (2017) concluded that, at landscape scales, fragmentation often has weak or even positive effects on biodiversity. Conversely, other studies contend that fragmentation can act at the patch scale, driving negative ecological outcomes generally (Chase et al., 2020; Fletcher Jr. et al., 2023; Gonçalves-Souza et al., 2025). Chase et al. (2020), while synthesizing the relative contributions of passive sampling and ecosystem decay, found that small habitat patches contain fewer species than expected based on patch area alone. These contrasting findings underscore the ongoing debate over the precise relationship between habitat fragmentation and biodiversity (Zhang et al., 2023a,b).

Ecological theory suggests multiple mechanisms through which fragmentation can exert either negative or positive impacts on biodiversity (Hanski et al., 2013; Thompson et al., 2017; Rybicki et al., 2020). For instance, negative outcomes often stem from harmful edge effects or minimum patch-size constraints, whereas positive effects may arise from increased habitat and resource diversity, beneficial edge influences, landscape complementation, reduced competition, or predator refugia (Fahrig, 2017; Fahrig et al., 2022; Valente et al., 2023a,b). However, many models derived from these mechanisms focus exclusively on taxonomic diversity—often measured as species richness—as the sole response variable (Fahrig, 2013; Püttker et al., 2020; Valente et al., 2023a,b; Yan et al., 2024; Gonçalves-Souza et al., 2025). Few studies explicitly assess how both individual species and entire communities respond to fragmentation while accounting for imperfect detection. Although species richness is a useful index for summarizing ecological patterns, it overlooks species identities and can fail to capture key community-level responses to fragmentation (Iknayan et al., 2014; Valente and Betts, 2019). Treating all species equally can also mask compositional and functional shifts that are crucial for conservation (Fletcher et al., 2018; Zhang et al., 2024). Moreover, richness-based metrics ignore differences in detection probability among species, potentially biasing or misrepresenting findings (Kéry et al., 2008; Zipkin et al., 2009). For example, certain species that are sensitive to fragmentation are simply harder to detect (Wang et al., 2015). If detection is not accounted for, common species may dominate richness estimates at the expense of cryptic or endangered species—often those of highest conservation concern (Valente and Betts, 2019). Gaining insight into species-specific responses to fragmentation is therefore fundamental to effective conservation planning, enabling better prioritization of vulnerable species (Zhang et al., 2024).

Understanding how interspecific variation in functional traits shapes species' vulnerability to habitat fragmentation is crucial for developing effective conservation strategies (Farneda et al., 2015; Wang et al., 2015). Species traits can reflect a species' capacity to cope with human-driven environmental change (Yu et al., 2024), and empirical studies highlight specific traits—such as dispersal ability, trophic level, body size, and niche breadth—that modulate a species' response to fragmentation (Hagen et al., 2012; Pfeifer et al., 2017; Betts et al., 2019). For instance, low mobility, high trophic level, and large body size are commonly associated with increased vulnerability (Pfeifer et al., 2017; Wang et al., 2024), whereas species with wider dietary or habitat niches often fare better (Hagen et al., 2012). Generalists may even thrive in fragmented landscapes by exploiting resources both within habitat patches and in the surrounding matrices (Chalfoun et al., 2002; Prugh et al., 2023; Tao et al., 2024). Consequently, multiple factors—including dispersal capacity, habitat specialization, diet, and body size—may need to be considered jointly to better predict species' responses to habitat fragmentation (Hagen et al., 2012). Trait-based approaches therefore provide more realistic forecasts of community-level impacts (Davies et al., 2004), and a comprehensive assessment of functional traits could help clarify the broader effects of fragmentation on biodiversity (Haddad et al., 2015). Moreover, habitat fragmentation affects functional traits related to persistence, activity patterns, dispersal, and resource use. These effects can lead to shifts in community functional diversity (Zambrano et al., 2019). For instance, as habitat structure homogenizes (Tscharntke et al., 2008; Suárez-Castro et al., 2020), the subsequent decline in functional diversity can impair key ecosystem processes, functions, and services (Zambrano et al., 2019). Although some evidence suggests these declines can be substantial, our understanding of exactly how fragmentation drives changes in functional diversity remains limited (Suárez-Castro et al., 2020).

Here, we investigated the associations between forest fragmentation (quantified as patch density (Riva et al., 2024)) and mammal species richness and community occupancy (i.e., average occupancy across the species community) as well as functional richness (measured as the hypervolume of functional space occupied by mammal communities (Mammola and Cardoso, 2020), while controlling for imperfect detection and human disturbance. We used an eight-month-long camera-trap dataset from 731 trapping stations and Bayesian hierarchical multispecies occupancy models to estimate community richness, occupancy, and functional diversity in the Hengduan Mountains of southwest China. Occupancy provides a reliable estimate of species distribution by accounting for observation errors and serves as a practical alternative to abundance estimation, which is often challenging (Efford and Dawson, 2012). Functional richness (the total variation in functional traits within a community) incorporates both species identity and the trait differences between them (Schleuter et al., 2010); as such, it offers a more mechanistic understanding of the ecological consequences of fragmentation than species richness alone (Jarzyna and Jetz, 2018). We hypothesized that increasing forest fragmentation reduces both community occupancy and functional richness. We posed two questions: (1) How does forest fragmentation relate to community occupancy and functional richness? (2) How do species' functional traits mediate responses of forest mammals to fragmentation? We predict lower community occupancy in highly fragmented forests because isolation among patches restricts movement and distribution ranges (Tucker et al., 2018). We also predict lower functional richness with increasing fragmentation as a result of biotic homogenization in highly fragmented landscape (Gámez-Virués et al., 2015). Our study aimed to provide novel insights into how functional traits shape mammalian responses to forest fragmentation, with broader implications for managing and conserving mammals and other functionally important wildlife groups in human-modified mountain forests.

2. Materials and methods

2.1. Study area

The Hengduan Mountains, situated within the Mountains of the Southwest China Biodiversity Hotspot (Fig. 1), encompass the Three Parallel Rivers UNESCO World Heritage Site (Xu et al., 2019). Renowned for its unique geology, complex topography, and high biodiversity, this region supports an exceptionally high diversity of

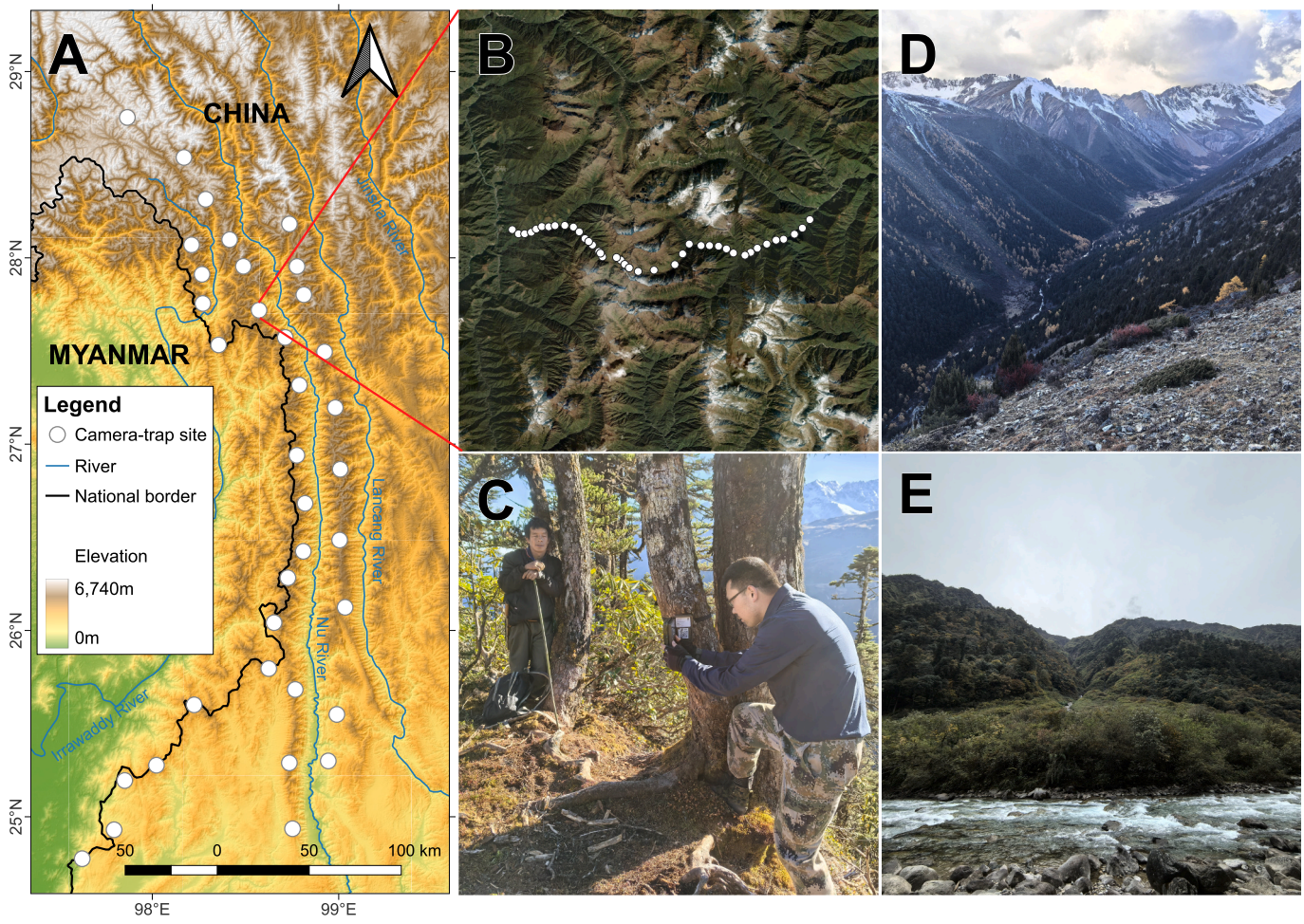


Fig. 1. Distribution of study locations in the Hengduan Mountains, southwestern China. Panel A shows the 36 sampling sites (white dots on map) across the region; Panel B provides a zoomed-in view of a representative site illustrating camera station spacing; Panel C shows an example camera trap deployment. Panel D and E show a representative landscape photographs from the study area.

species and ecological zones (Zhang et al., 2021). The climate of the Hengduan Mountains is predominantly influenced by the southwest monsoon originating from the Indian Ocean, fosters a wide range of vegetation types (Zhang et al., 2021). These vegetation zones transition along an elevational gradient, encompassing tropical monsoon forests (<1000 m), subtropical evergreen broad-leaf forests (1000–2600 m), temperate deciduous broad-leaf forests (1000–3000 m), sub-alpine coniferous forests (2700–3500 m), and alpine meadows (>3400 m) (Wang et al., 2007). The Hengduan Mountains serve as a critical habitat for several globally threatened mammals, including the clouded leopard (*Neofelis nebulosa*), dhole (*Cuon alpinus*), and takin (*Budorcas taxicolor*) (Li et al., 2024a,b). Human activities in the region are diverse, encompassing scattered settlements, smallholder cultivation and dispersed land management practices, livestock grazing, natural resource extraction, and transportation infrastructure (Li et al., 2022). The convergence of these anthropogenic pressures, alongside the region's remarkable biodiversity poses complex conservation challenges, making the Hengduan Mountains an important system for investigating the effects of fragmentation on wildlife diversity and distribution.

2.2. Camera trapping data

As part of a long-term monitoring program in the Hengduan Mountains, a key component of the China Biodiversity Monitoring and Research Network (Sino BON), we deployed 836 camera traps (Yianws L720) across 36 sites in mountain forests. We sampled forest sites spaced at least 20 km apart to represent the main vegetation types in the region. Our sampling sites covered a range of forest areas with different levels of human activity, proximity to villages, and degrees of forest fragmentation. Each site contained 20–30 camera-trapping stations, with the exact number determined by factors such as elevational range, forest area, fragmentation, and accessibility (Kays et al., 2020; Wearn and Glover-Kapfer, 2017). Camera stations were spaced at least 1 km apart and were not baited to avoid bias in wildlife behavior (Caravaggi et al., 2020). Camera traps were affixed to trees approximately 1 m above ground level and positioned away from trails frequently used by humans to reduce the risk of vandalism (Meek et al., 2016; Caravaggi et al., 2020). Camera trapping observations of mammals were collected during a single primary sampling period between October 2021 and May 2022. Camera sensitivity was set to low to minimize false triggers caused by non-animal movements. Each detection event was programmed to capture three photos, with a minimum interval of 3 s between successive detections (Palencia et al., 2021)). Photographs were identified to species level wherever possible, and human presence—including people, livestock, and dogs—was recorded to estimate human activity intensity at each station. To ensure data independence, multiple images of the same species (including humans) from the same camera were considered separate records only if they were at least 1 h apart (Niedballa et al., 2016; Wearn and Glover-Kapfer, 2017). Despite some camera stations failing to provide usable data due to malfunctions or loss, the total realized sampling effort amounted to 79,108 camera days from 731 operational camera traps.

2.3. Anthropogenic and habitat covariates

For each camera-trapping station, satellite-based covariates (patch density, human modification, annual mean temperature, percent forest cover, and altitude) were extracted using a 1-km buffer radius around the sampling station. A 1-km radius captures local-scale habitat changes and human activities that influence species occurrence and movement of medium- and large-bodied mammals relatively well (Li et al., 2024a,b). Fragmentation can be quantified using various metrics, from straightforward ones like patch density to more intricate measures such as the Edge Influence index (Martin et al., 2025). Patch density is defined as the number of distinct forest patches per 100 ha within the buffer area around each camera station; higher values indicate greater landscape fragmentation, reflecting subdivision of forest into more numerous, smaller units. To assess sensitivity to fragmentation metric choice, we computed four complementary landscape metrics within the 1-km buffer: patch density, edge density, mean patch size, and mean perimeter-area ratio (Hesselbarth et al., 2019). We extract the four metrics based on the ESA WorldCover dataset for 2021, which provides global land cover data at a 10-m resolution (https://viewer.esa-worldcover.org/worldcover) (Zanaga et al., 2022). All metrics were calculated at the landscape level using the "landscapemetrics" package (Hesselbarth et al., 2019) in R. The four metrics were standardized and incorporated individually into replicate occupancy and GAM analyses. Results were qualitatively consistent across metrics (Figs. S1–S3). Given its ecological interpretability and policy relevance (Martin et al., 2025; Riva et al., 2024), we retained patch density as the fragmentation indicator for further analysis. We use the term "forest fragmentation" throughout this study, as our metric (patch density) specifically quantifies the subdivision of forested areas. We acknowledge that for generalist species capable of utilizing the non-forested matrix, this represents forest—rather than absolute habitat—fragmentation. Where we use "habitat fragmentation", we follow the broader theoretical literature in which this term encompasses multiple landscape contexts. We included human modification and human presence around camera trapping stations to model how human activities potentially influence species diversity and distributions, as well as the trait diversity of mammal communities, which are reflected in occupancy patterns. Satellite-derived human modification data were obtained from the global human modification map (Kennedy et al., 2019), which offers a continuous 0–1 measure of the proportion of each 1-km^2 pixel directly modified by human activities, derived by mapping the physical footprint of 13 anthropogenic stressors, and then scaling each, and aggregating them. To assess human presence intensity at each camera-trapping station, we calculated relative abundance indices based on independent detections of domestic dogs, humans, and livestock, expressed as detections per 100 camera-trap days. Additionally, we measured habitat covariates that potentially influence species ecological functions and the trait diversity of mammal communities (Li et al., 2022). These covariates included annual mean temperature, which is a primary driver of community structure across taxa (Hawkins et al., 2003); percent forest cover, which provides food resources, thermal cover, and escape shelter (Li et al., 2022); and altitude. Climate variables were sourced from the WorldClim database (www.worldclim.org), while percent forest cover data were derived from the MODIS Vegetation Continuous Fields product (MOD44B) at a 250-m resolution for 2021. To meet the model assumptions of Gaussian data distributions for continuous variables, we applied a log transformation [log(x + 0.1)] to the patch density and human presence data. We then standardized all continuous variables (mean = 0, standard deviation = 1) to normalize the distributions and minimize the impact of outliers. We tested for multicollinearity among covariates (Table S1), applying a threshold of $|r| = 0.7$ (Dormann et al., 2013). We detected high correlations between annual mean temperature and altitude ($r = -0.79$) and between patch density and percent forest cover ($r = -0.82$); we subsequently excluded annual mean temperature and percent forest cover from multiple regression analysis (Smith et al., 2009). Variance inflation factors (VIFs) confirmed no severe multicollinearity (max VIF = 3.2). To evaluate how well our sampling sites represent the broader study region, we compared environmental variables at camera stations with regional distributions derived from landscape raster (Supplementary Table S2). Our sampling encompassed the Hengduan Mountains' core environmental gradients—including altitude (674–4306 m; covering 78.8% of the regional range), patch density (0.41–53.70; 85.1%), and elevation (674–4306 m; 81.1%).

2.4. Species trait data

Our study focused on medium- and large-bodied terrestrial mammals weighing more than 500 g, as these species typically exhibit large spatial and dietary requirements, high behavioral plasticity, and heightened sensitivity to human modification and habitat fragmentation (Li et al., 2022). Functional traits of the target species were compiled from a global mammal trait database and a national mammal database (Cooke et al., 2019; Ding et al., 2022). For each species, we extracted data on four functional traits—body mass, habitat width, diet, and activity period (Table S3)—that are widely recognized in ecological studies as strong indicators of functional diversity (Gorczynski et al., 2023). A substantial body of research supports the habitat width–elevational or vertical range size association (e.g., Pintor et al., 2015; Lee-Yaw et al., 2016). We used log of elevational range size to estimate habitat width for each species (Pintor et al., 2015). We used three diet categories (carnivore/herbivore/omnivore) following Atwood et al. (2020) and to ensure sufficient sample size per category for Gower distance calculation. Finer splits (e.g., insectivore, frugivore) would leave >50% of species in sparse categories, violating assumptions of hypervolume estimation (Mammola and Cardoso, 2020). Thus, species consuming ≥90% animal-based diet were categorized as carnivores, ≥90% plant-based diet as herbivores, and <90% animal- or plant-based diet as omnivores (Table S3). Species activity periods were categorized into crepuscular, diurnal, nocturnal and cathemeral (Ding et al., 2022). These traits capture key aspects of species' ecological requirements and effectively evaluate their responses to habitat fragmentation and environmental changes (Hagen et al., 2012; Gorczynski et al., 2023). For species with missing trait data, trait values were imputed based on the traits of closely related species within the same genus, ensuring consistency across taxa. Trait values were imputed for 39 species across 4 traits (body mass, habitat width, activity pattern and diet, Table S3). To enhance comparability among traits with different value ranges, body mass data were log-transformed.

2.5. Occupancy modeling

We employed Bayesian hierarchical occupancy modeling (Dorazio and Royle, 2005) to investigate the effects of forest fragmentation on community-level occupancy and individual species occurrence, while explicitly accounting for imperfect detection (MacKenzie et al., 2002). Species detected at fewer than 10 independent camera stations were excluded from occupancy modeling, leaving 39 species for analysis (Table S4). The modeling framework consisted of two sub-models: (1) the state process model, which describes the ecological system, and (2) the observation process model, which accounts for the sampling protocol (MacKenzie et al., 2002). Accounting for imperfect detection enabled us to estimate the probability of a mammalian species being present at a station even if it was not directly observed. Each camera-trapping station was treated as a sampling unit, with 5 consecutive trap days considered as sampling occasions. A 5-day sampling occasion strikes a balance between model stability and ensuring an adequate number of repeat visits to each site (Burton et al., 2015; Li et al., 2018). Species were recorded as either detected or not detected at each station for each sampling occasion during which the camera was operational. We included a site-level random effect in our occupancy model by defining $w_{i,s}$ as a latent binary variable that indicate the presence and absence of

species i in site s, $w_{i,s}$ is governed by the hyperparameter Ω_s:

$$w_{i,s} \sim Bernouli(\Omega_s)$$

$$\Omega_s \sim Normal(\mu.s, \sigma.s)$$

where

$$\mu.s \sim Normal(0, 1000) \text{ and } \sigma.s \sim Uniform(0, 10)$$

We then modeled the occurrence ($z_{i,j}$) of species i at camera station j in site s as a Bernoulli random variable, $z_{i,j} \sim Bernouli(\psi_{i,j} \times w_{i,s})$, where $\psi_{i,j}$ represents the latent occupancy probability. Occupancy modeling typically relies on the key assumption of site closure (MacKenzie et al., 2002). However, this assumption can be relaxed when the home range of the target species exceeds the effective sampling area of camera traps (Efford and Dawson, 2012; Burton et al., 2015). Consequently, we interpreted the occupancy parameter ($\psi_{i,j}$) as an estimate of the probability that the species utilized the area surrounding the camera trap station, rather than true occupancy (Burton et al., 2015). Detection of species i at station j on occasion k conditional on station j being occupied was modeled as $y_{i,j} \sim Binomial\left(k_j, p_{i,j} \times z_{i,j}\right)$, where $p_{i,j}$ is the detection probability of species i at station j, given that i is present ($z_{i,j} = 1$), k_j is the total number of sampling occasions that station was operating (MacKenzie et al., 2006).

The occupancy model incorporated species-specific estimates for occupancy, treated as random effects drawn from a community-level hyper-parameter shared across all species. We define community occupancy as the mean posterior occupancy probability across all mammalian species detected at least 10 times, calculated at the camera station level. This metric reflects the central tendency of site-use probability for the entire community and serves as a community-level response variable (Greco et al., 2025). Random effects included intercepts for occupancy and detection, as well as slopes for four occupancy covariates: patch density, human modification, human presence, and altitude. We decided to fit a linear model after visually inspecting the residual vs. fitted plot of the model (Fig. S4). Detection probability was modeled with an intercept-only structure, as no environmental covariates showed significant effects on species detection probabilities (Table S5). For each random slope, the species-specific beta coefficient (βi) was modeled as $\beta_i \sim Normal(\mu.\beta, \sigma.\beta)$, where $\mu.\beta$ and $\sigma.\beta$ are the estimated mean and variance across all species. Vague priors were assigned to these terms, drawn from normal distributions with mean zero and variance 1000. To ensure model stability and convergence, only mammal species detected more than 10 times were included (Table S4). The occupancy models were implemented within a Bayesian framework using JAGS via the *R2jags* package (Su et al., 2024) in R version 4.3.2 (Appendix S1). We ran three Markov chain Monte Carlo (MCMC) chains for 20,000 iterations, discarding the first 12,000 as burn-in and retaining every 20th sample. We assessed convergence by visually inspecting the chains and by ensuring the Gelman-Rubin statistic for each parameter was close to 1 (Fig. S5). Beta coefficients for each covariate were presented as the mean of the posterior distribution, accompanied by 95% Bayesian credible intervals (BCIs) to quantify uncertainty (Table S6). Effects were considered significant if the 95% BCI did not include zero.

During each iteration, the model generates an occurrence matrix of camera station and species-specific z values. To estimate species richness at camera station j, we summed the number of estimated species (i.e., instances where $z_{i,j} = 1$) during iteration x. We then repeated this process for each of the 1200 iterations and used these values to generate a probability distribution representing camera station-specific species richness (Zipkin et al., 2010).

We further assessed occupancy-model sensitivity to the spatial scale of patch-density measurements by repeating analyses across buffer radii (250-m, 500-m, and 2-km). Results remained qualitatively consistent (Table S7), demonstrating robustness to scale variation. The 1-km scale yielded the strongest, most stable effects – exhibiting the lowest Deviance Information Criterion (DIC) and highest effect sizes in site-occupancy models – matching established patterns for medium-to-large terrestrial mammals (Li et al., 2024a,b; Burton et al., 2015).

2.6. Trait-based analysis

We estimated functional richness at each camera-trapping station by deriving station-level distributions from the occupancy model outputs, providing mean and standard deviation values for functional diversity estimates. Functional richness values were computed from the binary species occurrence matrices (z-matrices) generated during each of the 1200 retained MCMC iterations. For each iteration, species present at a given station ($z = 1$) were used to define the local community, and the convex-hull functional richness was calculated for that community. The mean and standard deviation of these 1200 estimates constituted the station-level functional richness values used in subsequent analyses. Functional richness was evaluated using hyperdimensional functional trait spaces constructed from probabilistic hypervolumes (Mammola and Cardoso, 2020). To create the global trait space, we calculated pairwise functional distances among all species included in the occupancy model (detected more than 10 times) based on their traits. Gower's distance metric was employed for the distance calculation as it is particularly suitable for datasets containing both numerical and categorical variables, standardizing these appropriately (Coutant et al., 2023). The resulting dissimilarity matrix was projected into a multidimensional trait space using principal coordinate analysis (PCoA) with Cailliez correction to address any negative eigenvalues (Soares et al., 2024). To represent the global trait space, we retained the first four PCoA axes, which collectively explained 89% of the total variation. Using this global trait space, functional richness was calculated for each station using the *hull.alpha* function with the "BAT" package in R, which scales convex-hull volume to 0–1 (Mammola and Cardoso, 2020).

Although species and functional richness was derived from the posterior samples of species-level occupancy for each site from the hierarchical models, we tested whether fragmentation further explained residual variance in species and functional richness. To assess the effects of fragmentation on species and functional richness, we modeled the species and functional richness for each camera-trapping station as response variables using generalized additive models (GAMs). The R package "mgcv" (Wood, 2023) was used to fit these models to the estimated species and functional richness. This approach revealed community-level processes not captured by species-specific responses. To explore the relationship between mean effects of patch density on individual species occupancy and species-specific ecological traits, we performed backward linear model selection using the "MASS" R package (Ripley et al., 2025). This approach aimed to determine whether species with certain functional traits exhibited stronger responses to the patch density. All functional traits—body mass, habitat width, diet, and activity period—were included as predictors, with species-specific regression coefficients for occupancy serving as the response variable, weighted by the inverse of the squared standard deviation. Finally, we ran backward linear model selection to identify the most parsimonious model to determine whether species with particular traits were more vulnerable to patch density. In the backward model selection, diet was included as a categorical variable with carnivore as the reference category; reported coefficients for herbivore and omnivore therefore represent deviations from the carnivore baseline.

2.7. Structural equation modeling

Tests of fragmentation effects are often confounded by landscape composition due to the high collinearity between habitat amount and configuration (e.g., Smith et al., 2009; Ruffell et al., 2016). This collinearity's causal basis should be explicitly incorporated into statistical models by partitioning the direct versus indirect pathways through

which fragmentation influences ecological responses, mediated by habitat loss (Ruffell et al., 2016). The pronounced negative correlation ($r = -0.82$) between patch density, which serves as our metric for habitat fragmentation, and forest cover introduces significant collinearity. To mitigate this issue in our hierarchical occupancy models, we excluded forest cover, adhering to established statistical guidelines to prevent multicollinearity, as it can otherwise inflate variance and bias the estimation of coefficients (Smith et al., 2009). However, as multiple regression does not inherently account for indirect effects resulting from collinearity between habitat fragmentation and forest cover, we further employed a piecewise structural equation model. Specifically, our approach incorporated two linear mixed-effects models (LMMs): one set modeled species and functional richness, while the other modeled habitat fragmentation. The richness models included patch density and forest cover as predictors, whereas the fragmentation models used forest cover and human modification as explanatory variables. To account for potential dependencies within our data arising from spatial autocorrelation, sampling periods, species pools, and landscape dynamics, camera trapping sites were treated as random intercepts in all LMMs. In real landscapes, the causal relationship between habitat fragmentation and forest cover remains uncertain (Ruffell et al., 2016). Given the difficulty in determining whether fragmentation drives forest loss or vice versa, we tested alternative path models with fragmentation predicting forest cover. For easier comparison, we combined both types of path analyses in the final path diagram. These path analyses allowed us to quantify and distinguish both the direct and indirect influences (mediated through forest cover) of habitat fragmentation on species and functional richness (Ruffell et al., 2016).

3. Results

A total of 39 species, each detected more than 10 independent stations, were included in this analysis (Table S4). The species exhibited a wide range of average body masses, from 0.51 kg (*Mustela kathiah*) to 302 kg (*Budorcas taxicolor*). Environmental conditions at the camera trap stations varied substantially: patch density ranged from 0.41 to 53.70 (Fig. S6), human modification values spanned from 0.02 to 0.62, and human presence ranged from 0 to 118.66. Altitude across the

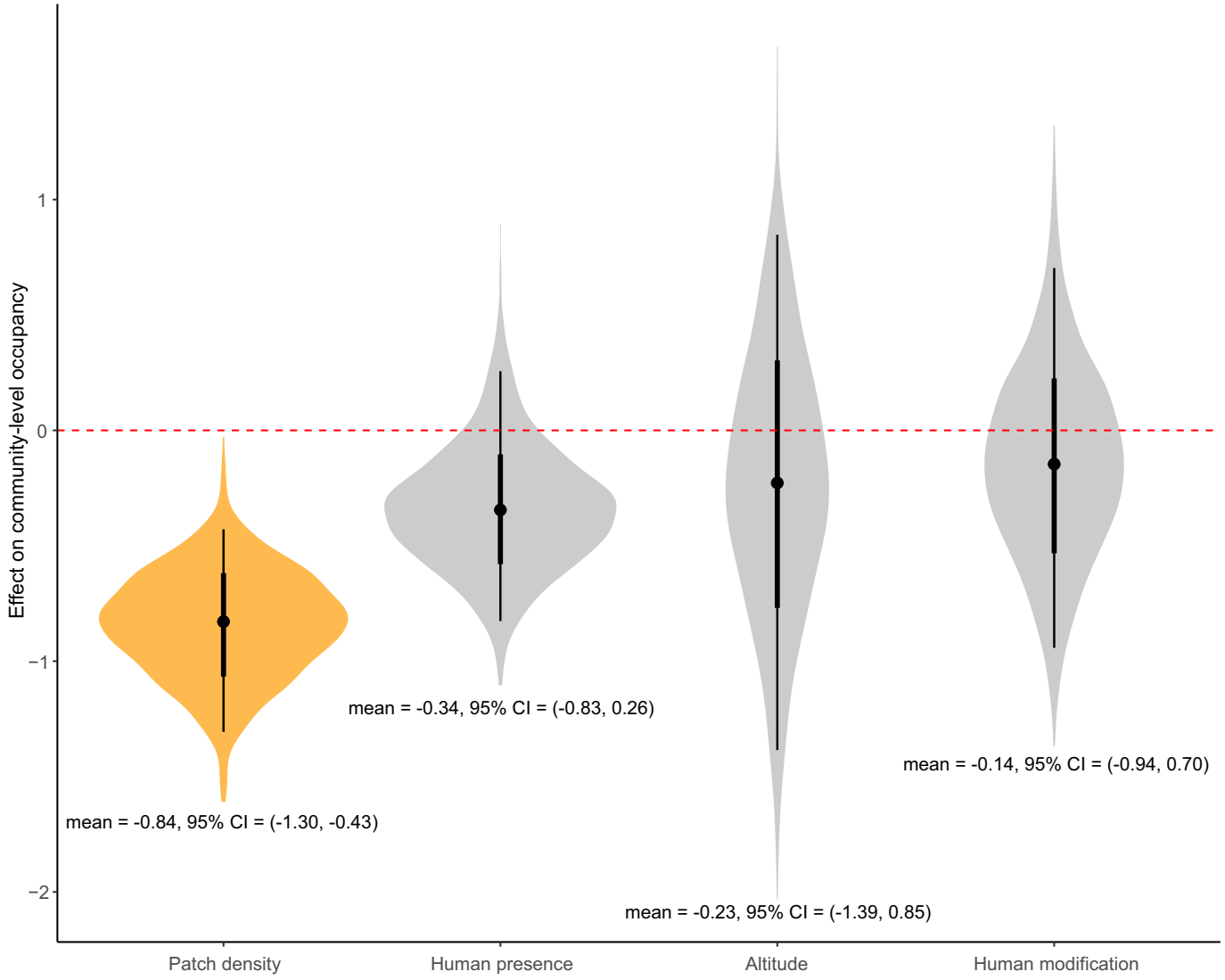


Fig. 2. Beta coefficient plot illustrating the relative effect sizes of standardized habitat covariates on mean mammalian species occupancy, based on camera trapping survey data from the Hengduan Mountains, southwest China. Points represent the mean effect sizes, with thin bars indicating 95% Bayesian credible intervals and thick bars indicating 90% Bayesian credible intervals. The color of the distributions reflects whether the 95% credible interval includes zero: yellow indicates that the 95% credible interval does not include zero, while grey indicates that it does. The 95% credible interval that excludes zero shows patch density having a significant negative relationship with mean mammalian species occupancy. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

stations extended from 674 to 4306 m. Community mean occupancy ranged between 0.12 and 0.57, while mean functional richness were between 0.26 and 0.89.

The results from the Bayesian hierarchical multispecies occupancy model revealed a significant negative association between patch density and mammalian species occupancy at the community level, as the 95% Bayesian credible interval excluded zero (Fig. 2). Both human presence and modification showed weak, nonsignificant negative associations with occupancy.

Species richness and functional richness were moderately correlated ($r = 0.431$, P-value < 0.001, Fig. S7). Generalized additive model identified a nonlinear relationship between estimated species richness and patch density. Species richness initially showed a sharp decline with increasing patch density and then leveled out at patch density of about 20, followed by a slight rebound at higher patch density (Fig. 3a). Functional richness exhibited distinct relationships with patch density. Overall, functional richness showed a nearly linear decrease with increasing patch density (Fig. 3b).

The piecewise structural equation models demonstrated robust model fits (species richness: Fisher's C = 2.56, P-value = 0.28; functional richness: Fisher's C = 4.39, P-value = 0.16). Notably, standardized path coefficients consistently revealed habitat fragmentation as a stronger determinant of both species and functional richness compared to forest cover (Fig. 4a). The effect of habitat fragmentation on species richness ($\beta = -0.605$) substantially exceeded the influence of forest cover ($\beta = -0.038$). Similarly, habitat fragmentation exerted nearly three times greater influence ($\beta = -0.252$) on functional richness than forest cover ($\beta = 0.093$). Nevertheless, the direct influence of forest cover on functional richness remained statistically significant even when controlling for habitat fragmentation (Fig. 4b).

Occupancy for most species (33 out of 39, 85%) had a negative relationship with patch density (Fig. 5), although the mean effect size varied among species. Notably, occupancy for 14 out of 39 species (36%) showed significant and negative correlations with patch density (Fig. 5). In contrast, one species, i.e., *Macaca mulatta* showed significant and positive correlations with patch density (mean coefficient = 0.94, 95% BCI = 0.46 to 1.47) (Fig. 5).

Backward stepwise model selection identified habitat width, herbivore diet, and nocturnal activity as the key predictors of species responses to patch density in the most parsimonious model. The most parsimonious model revealed habitat width had a significant positive effect on species response to patch density (mean coefficient = 0.75; 95% credible interval = 0.42 to 1.06), while nocturnality exhibited a significant negative effect (mean coefficient = −0.53; 95% credible

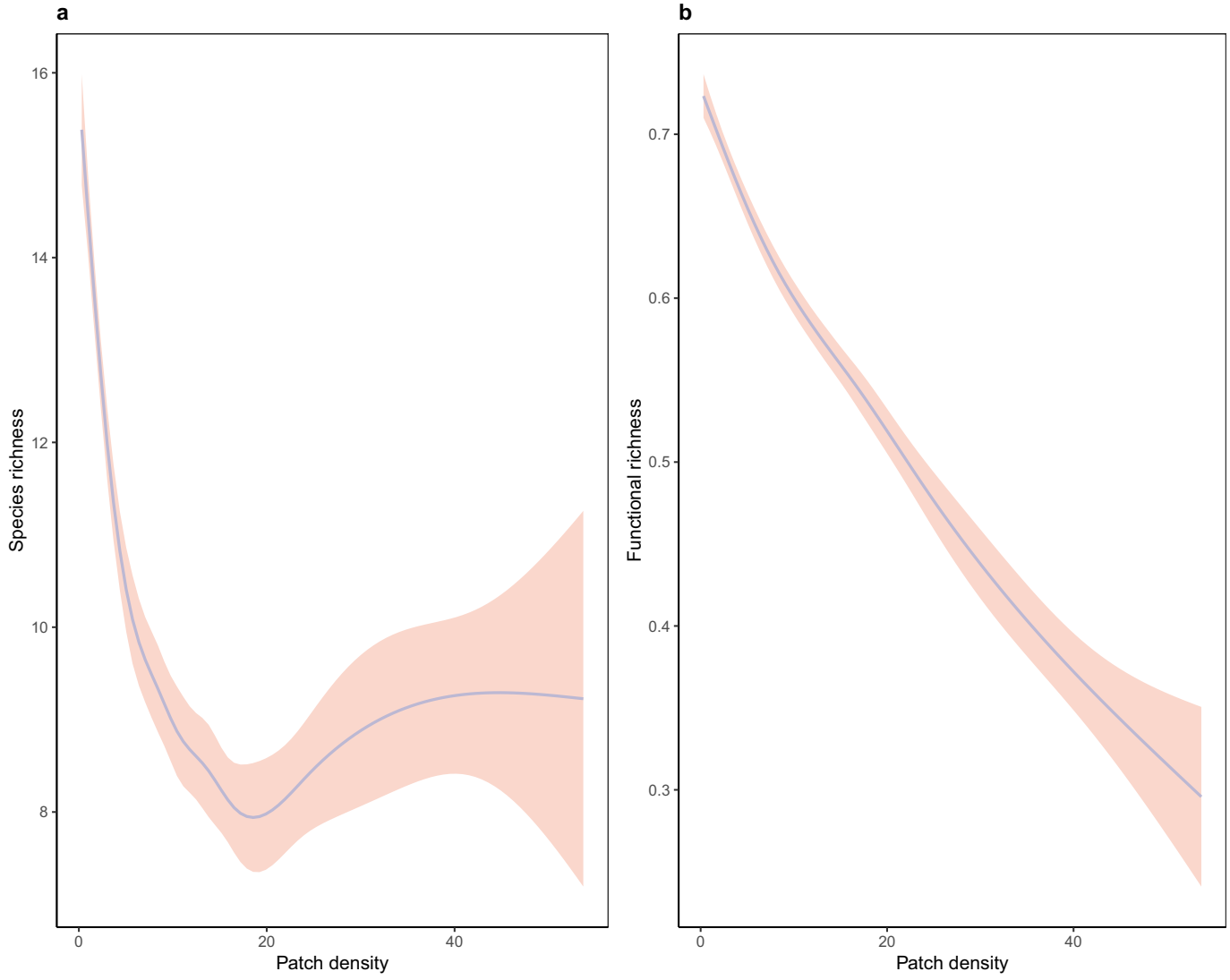


Fig. 3. Relationships between standardized fragmentation metrics (patch density) and mean mammalian species richness and functional richness. Blue lines indicate the regressions' estimated relationship between the predictor and response variables, with the shaded areas indicating the 95% credible interval of the predicted values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

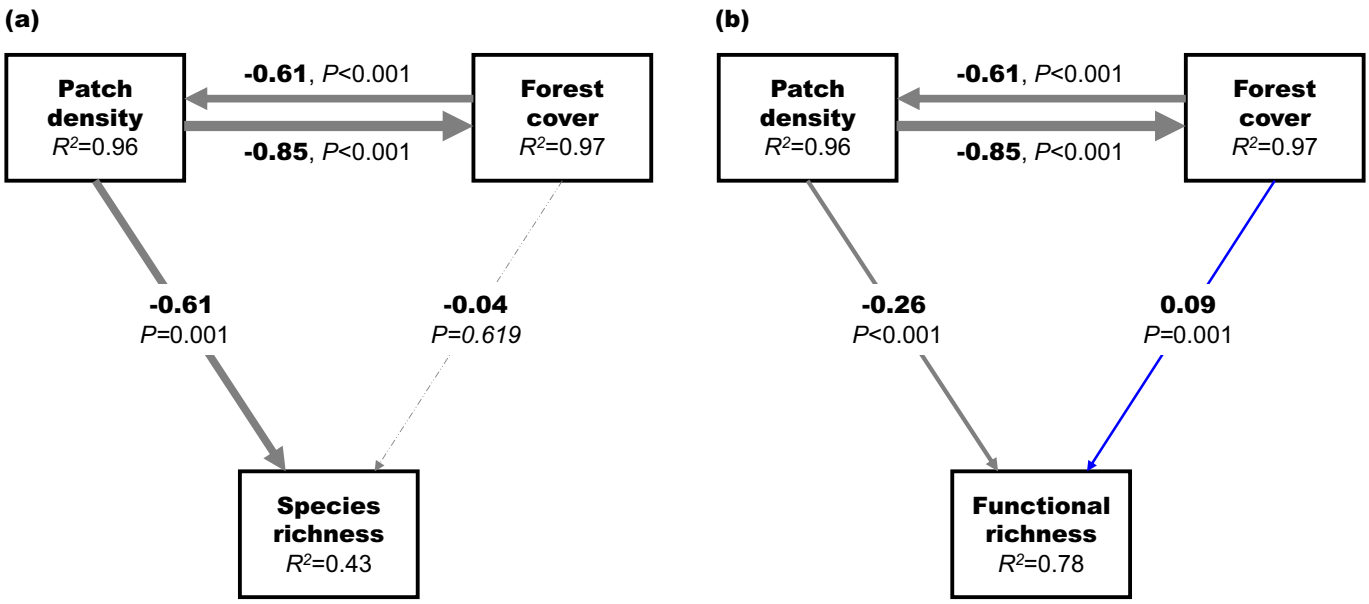


Fig. 4. Path diagram illustrating the path model examining how habitat fragmentation, measured as patch density, and habitat amount, measured as forest cover, affect species richness (a) and functional richness (b). Arrows indicate unidirectional relationships among variables. Each diagram summarizes two models: one where fragmentation is the exogenous variable and habitat amount is endogenous, and vice versa. Colored arrows show significant positive (blue) or negative (grey) relationships ($p < 0.05$), while grey dashed lines indicate non-significant paths. Standardized path coefficients are shown on the lines corresponding *P*-values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

interval = −0.98 to −0.08) (Fig. 6).

4. Discussion

Based on in situ observational data from an intensive camera-trapping survey spanning a broad gradient of forest fragmentation in the Hengduan Mountains, southwest China, we evaluated how forest fragmentation affects multiple diversity facets among medium- and large-sized mammals, taking imperfect detection into account. According to the standardized path coefficients, habitat fragmentation consistently emerged as a more significant driver of species and functional richness than forest cover (Fig. 4), supporting the habitat patch hypothesis in forest landscapes, which predicts a negative relationship between fragmentation and biodiversity. In particular, functional richness continually decreased even when species richness leveled out along gradients of patch density (Fig. 3b), indicating the decoupling effects of fragmentation on species and functional richness. Additionally, we observed species-specific responses, with nocturnal mammals being notably less likely to occupy habitats characterized by higher forest patch density (Fig. 6). This underscores the importance of considering ecological traits, such as activity patterns, in assessing vulnerability to habitat disturbances. Moreover, we demonstrate that even large, highly mobile species can be adversely affected by the loss of contiguous forest cover, emphasizing the need for targeted conservation strategies in fragmented landscapes. Forest fragmentation continues to pose critical challenges by eroding biodiversity and ecological processes (Pfeifer et al., 2017). While many studies have focused on broad, summed metrics such as species richness, these approaches can obscure important differences among species with distinct habitat requirements and may underestimate the true effects of fragmentation (Hanski, 2015). Here, we applied hierarchical multispecies occupancy models that account for imperfect detection, enabling us to discern individual species- and community-level responses to forest fragmentation. This framework offers a more precise understanding of how taxonomic and functional richness patterns shift under fragmentation pressure. By integrating trait-based approaches, we further highlight the specific characteristics that make certain mammals more susceptible to forest fragmentation, facilitating the development of more effective conservation and management plans. In doing so, we advance understanding of the fragmentation–diversity relationship and offer practical guidance for mitigating biodiversity loss in rapidly changing landscapes.

4.1. Effects of forest fragmentation on occupancy patterns

Occupancy is a metric of diversity that accounts for imperfect detection and is a useful measure to overcome the inherent difficulties of abundance estimation (Steenweg et al., 2018). Community occupancy represents the central tendency of site use probability for all species belonging to a community and reflects site-level responses of wildlife to local habitat changes (Greco et al., 2025). Overall, the observed community-level response to patch density indicates that mean site use probability declines in more fragmented areas. Declines in community occupancy would ultimately affect species richness when a species no longer occupies an area (Greco et al., 2025). More sensitive species may disappear over time under increasing fragmentation (Betts et al., 2019). Similarly, global assessments have shown reduced mammal distributions with higher fragmentation (Crooks et al., 2017). These findings align with habitat patch hypothesis, which posits patch configuration as a key determinant of species presence in fragmented habitats (Hanski et al., 2013). Although debate persists regarding whether fragmentation (Fahrig, 2017) exerts a net negative or positive effect on biodiversity (Fletcher et al., 2018; Fahrig et al., 2019), a substantial body of work suggests that fragmentation can contribute to local species extinctions (Haddad et al., 2015; Fletcher et al., 2018; Chase et al., 2020).

Habitat fragmentation frequently disrupts the spatial distribution of resources, altering movement conditions and occupancy patterns, and ultimately affecting individual fitness (Crooks et al., 2017). Identifying which species or communities are most vulnerable to fragmentation is a critical step toward effective conservation planning (Wang et al., 2015, 2024), particularly in light of continued pressures on forest habitats worldwide. At the species level, effect sizes for patch density were negative for 33 of the 39 species analyzed (Fig. 5, Table S6), indicating an overall decline in occupancy along fragmentation gradients. Nevertheless, our results also demonstrate variability in species responses, reflecting different habitat requirements (Betts et al., 2022) and offering insight into why the role of fragmentation relative to habitat loss

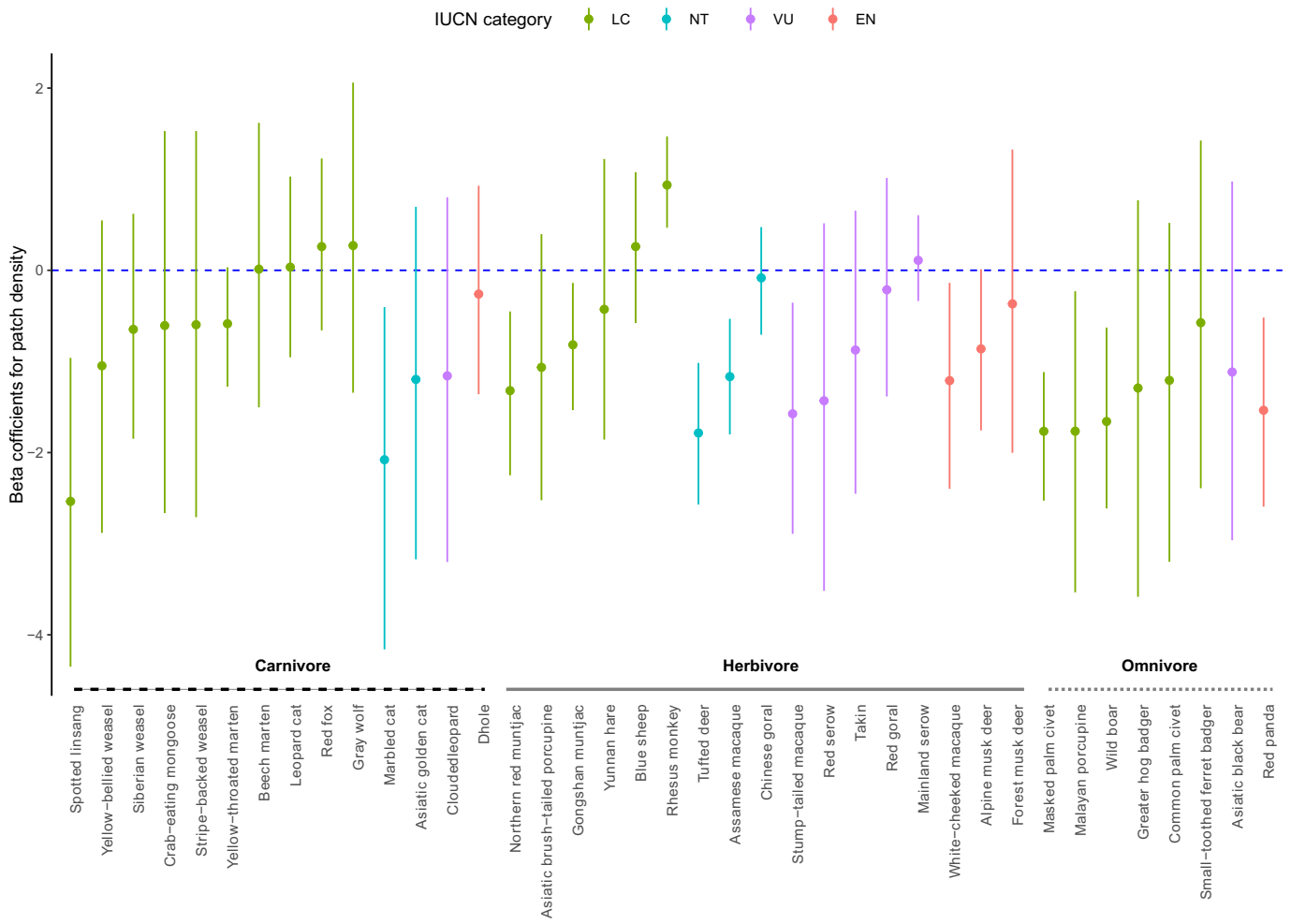


Fig. 5. Mean effect sizes of patch density on species-specific occupancy, with species ordered first by diet group, and then by IUCN Red List category (LC, NT, VU, EN) and mean effect size within each category. Points represent the mean effect sizes, with vertical lines indicating 95% Bayesian credible intervals. Colors of point and line correspond to IUCN Red List categories: LC (Least Concern), NT (Near Threatened), VU (Vulnerable), and EN (Endangered). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

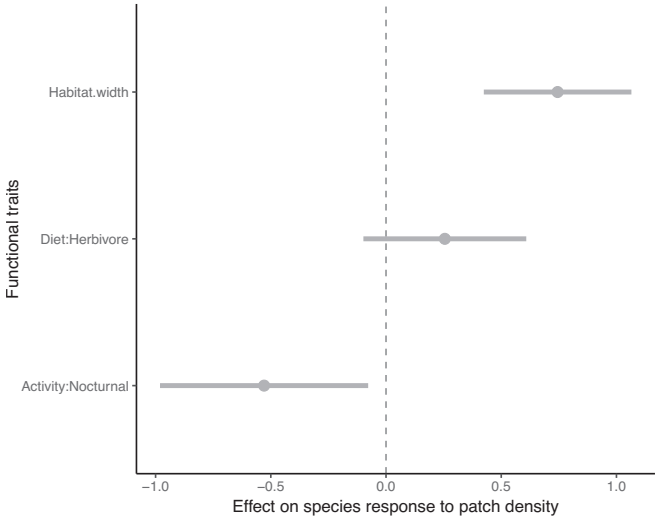


Fig. 6. Associations between ecological traits and species responses to habitat covariates, as identified through backward stepwise model selection. In the backward model selection, diet was included as a categorical variable with carnivore as the reference category; reported coefficients for herbivore and omnivore therefore represent deviations from the carnivore baseline.

remains a subject of ongoing debate (Kuipers et al., 2021). Based on the mean coefficients of patch density on mammalian species occupancy, seven species (Rhesus monkey, dhole, blue sheep, red fox, mainland serow, spotted linsang, and beech marten) showed a positive relationship with patch density (Table S6), indicating that the occurrence probability of these species may increase with increasing fragmentation.

Notably, the most imperiled species on the IUCN Red List (Endangered and Vulnerable) were not only the most vulnerable to fragmentation (with 10/11 of them showing a significant negative effect) but also included the species with the steepest declines in occupancy (e.g., the endangered Chinese red panda). This finding underscores a severe conservation concern: the very species that are already at the highest risk of extinction are also those most sensitive to ongoing habitat fragmentation. This creates a dangerous feedback loop, where fragmentation disproportionately impacts the most vulnerable, further accelerating their decline. Our results reinforce the urgent need for targeted, risk-based conservation intervention.

4.2. *Effects of forest fragmentation on species richness*

Habitat loss is often the primary driver of isolation and fragmentation (Fletcher et al., 2018), making it challenging to disentangle the independent impacts of fragmentation on biodiversity (Fletcher et al., 2018; Püttker et al., 2020). By accounting for the causal basis of collinearity between habitat fragmentation and forest cover in our

piecewise structural equation models, we were able to estimate the independent effect of each variable while controlling for the other. Even after accounting for forest cover, a proxy for habitat amount, our results revealed a significant negative effect of fragmentation on species richness (Fig. 4a), underscoring the importance of habitat configuration in determining occurrence patterns of forest-dwelling mammals. Thus, our study contributes to a growing consensus that local-scale fragmentation can be detrimental to species that rely on extensive and contiguous forest cover (Kuipers et al., 2021).

We observed a nonlinear relationship between species richness and forest fragmentation (Fig. 3a), indicating a complex response of richness to fragmentation. Species richness initially declined with increasing patch density and then leveled out beyond a threshold value of 20 (Fig. 3a). Forest fragmentation may filter forest-dependent species, reducing species richness along fragmentation gradients (Fletcher et al., 2018). Moreover, fragmentation may create empty niches for generalist animals (Semper-Pascual et al., 2023). After a certain threshold of fragmentation (at patch density of about 20), the local disappearance of sensitive species may be compensated for by the colonization of new adaptable species (Dornelas et al., 2014; Greco et al., 2025). In highly fragmented landscapes, the loss of forest-dependent species is often followed by the influx of habitat generalists that readily colonize the newly vacated patches (Dornelas et al., 2014). As a result, species richness typically declines as fragmentation increases until a threshold level beyond which the relationship may plateau or even shift to a positive one. Such relationship may be a consequence of local extinction filtering after a certain level of fragmentation, i.e., reduced distributions of the most vulnerable species and thriving of the more resilient species (Balmford, 1996).

4.3. Effects of forest fragmentation on functional richness

Species functional traits determine which species can persist under increasing fragmentation (Reich et al., 1997); because functional richness is calculated from the traits of co-occurring species, the selective filtering of species with particular traits—specifically habitat specialists and nocturnal species—provides the mechanistic link between fragmentation and functional richness loss (Hagen et al., 2012). However, few studies have investigated how forest fragmentation influences the functional traits and diversity patterns of medium- and large-sized mammals, leaving the generality of these effects unclear. Our findings indicate that forest fragmentation exerts a significant negative impact on mammalian functional richness. In particular, functional richness declined sharply with increasing patch density (Fig. 3b), suggesting that fragmentation not only reduces species distributions but can also cause substantial losses in functional richness—a trend that may produce cascading effects on community interactions and ecosystem functioning (Gorczynski et al., 2021; Li et al., 2022). Notably, functional richness continually decreased even when species richness leveled out along gradients of patch density (Fig. 3b), indicating that functional trait diversity is more vulnerable than species richness in fragmented forests. Although forest cover showed a significant and positive effect on functional richness, patch density was identified as the dominant driver for functional richness patterns across the forests of southwest China (Fig. 4b). Our results suggest that forest fragmentation is filtering for species that are more functionally similar, thus homogenizing functional diversity. Because this reduction in functional richness can compromise key ecosystem processes (Farneda et al., 2015), we advocate shifting the focus of fragmentation-diversity relationship to elucidating the effects of fragmentation on community structure and functional diversity simultaneously (Zambrano et al., 2019).

4.4. Key functional traits in relation to fragmentation and human disturbance

Functional traits are increasingly recognized as key predictors of species' responses to habitat fragmentation (Farneda et al., 2015; Keinath et al., 2017; Pfeifer et al., 2017), in part because they can reconcile seemingly inconsistent or contrasting findings across studies (Ewers and Didham, 2006). An ongoing challenge in modeling fragmentation-driven biodiversity changes is pinpointing those traits most strongly linked to species persistence. Our results indicate that mammal species with narrower habitat width or nocturnal behavior were more adversely affected by fragmentation (Fig. 6). These patterns align with previous findings that highlight habitat width as a critical trait in determining sensitivity to fragmentation (Hagen et al., 2012; Chen et al., 2025), given that generalists can sometimes exploit resources in both the fragment and the surrounding matrix (Andrén, 1994; Hagen et al., 2012). Many nocturnal mammals rely on darkness for foraging, predator avoidance, and navigation (Shier et al., 2020). Human-induced modification of nighttime light levels in highly fragmented habitats can create a "landscape of fear", posing particular threats to shy or cryptic wildlife (Ortiz-Jimenez et al., 2025). Our findings reveal a critical conservation signal: species with narrow habitat breadth and nocturnal activity patterns are disproportionately vulnerable to fragmentation, suggesting that current protection strategies may be failing to safeguard forest-dwelling specialists. In particular, nocturnal species—often cryptic and poorly monitored—show the strongest negative responses, highlighting a significant blind spot in monitoring and management efforts. Given that 85% of species in our study exhibited negative occupancy responses, with 36% showing statistically significant declines, priority should be assigned to those with high extinction risk and low adaptive capacity. Taken together, our findings suggest that forest fragmentation selectively filter mammalian species based on specific traits, thereby shaping local community composition and functional diversity.

5. Conclusion

In summary, our study suggests that effective conservation strategies addressing habitat loss and fragmentation should focus on relatively local scales, where fragmentation exerts pronounced impacts on taxonomic and functional richness. By integrating a hierarchical occupancy, structural equation modeling and trait-based approaches, we provide a detailed framework for understanding how fragmentation influences multiple dimensions of biodiversity. Our camera-trapping survey primarily targeted regional, ground-dwelling mammals and therefore did not encompass the full forest biome or its vertical stratification. Occurrence data for marginally distributed and arboreal species were too sparse for robust statistical analysis. We excluded taxa with fewer than ten detections (Table S4) because, when observations are limited, changes in occupancy and detection probability cannot be reliably distinguished. While this filtering step is common in ecological modeling to reduce noise and improve parameter estimation (e.g., Gorczynski et al., 2023), it inevitably biases results toward the fragmentation responses of ground-dwelling generalists and may underestimate declines in rare or arboreal species that are particularly vulnerable to habitat loss. Nonetheless, the hierarchical occupancy framework still enables us to detect both species- and community-level responses for most medium- and large-bodied mammals to forest fragmentation. In general, mammalian communities had higher occupancy rates in less fragmented forests, and larger forest patches sustained greater functional richness. The correlation between functional vulnerability—such as niche specialization and nocturnality—and elevated extinction risk underscores an urgent need for species-specific conservation planning informed by functional traits and extinction risk status. These results also have implications for the single-large-or-several-small (SLOSS) debate, emphasizing that retaining large, contiguous forest areas in mountain landscapes is critical for supporting diverse mammalian communities. Our finding that nocturnal species and habitat specialists are disproportionately affected by forest fragmentation directly informs protected area design: conservation planning must prioritize

maintaining large, contiguous forest blocks to minimize edge effects and light pollution in the Hengduan Mountains. Concurrently, targeted monitoring programs for IUCN-listed species exhibiting the strongest negative responses to patch density—particularly the endangered Chinese red panda (*Ailurus styani*) and alpine musk deer (*Moschus chrysogaster*)—should be implemented immediately.

CRediT authorship contribution statement

Xueyou Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wenqiang Hu:** Software, Investigation, Data curation. **Kenneth Otieno Onditi:** Writing – review & editing, Visualization, Data curation. **Zhechang Hu:** Investigation, Formal analysis, Data curation. **Yang Liu:** Investigation, Formal analysis, Data curation. **Hongjiao Wang:** Investigation, Data curation. **Xuelong Jiang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

No AI tools were used to generate or analyze data or results. Language editing and grammar checking were assisted by Grammarly and Word Editor, with all scientific content verified and revised by the authors.

Declaration of competing interest

The work is all original research carried out by the authors. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. All authors agree with the contents of the manuscript and its submission to the journal. No part of the research has been published in any form elsewhere, unless it is fully acknowledged in the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biocon.2026.111828.

Data availability

The data and R scripts for this study can be accessed via Dryad using the following link: http://datadryad.org/share/LINK_NOT_FOR_PUBLICATION/w1CrpcBzR78dVHzcr_bskC_jqfQfIAZoie8JmvhNzDc. This resource will be made publicly available from the date of publication of this manuscript.

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